Creation of Amazon Redshift Serverless (Workgroup and Namespace)

Summary of the Lab

In this lab, Amazon Redshift Serverless is created without provisioning any clusters or nodes. A **namespace** and a **workgroup** are configured to manage compute and storage resources. The lab demonstrates how Redshift Serverless enables resource allocation and segregation using workgroups and namespaces. The workgroup capacity is configured using Redshift Processing Units (RPUs), and networking settings are updated to allow public access. Finally, inbound rules are configured in the security group to allow connections on the Redshift port (5439), enabling external tools and pipelines to connect to Redshift.

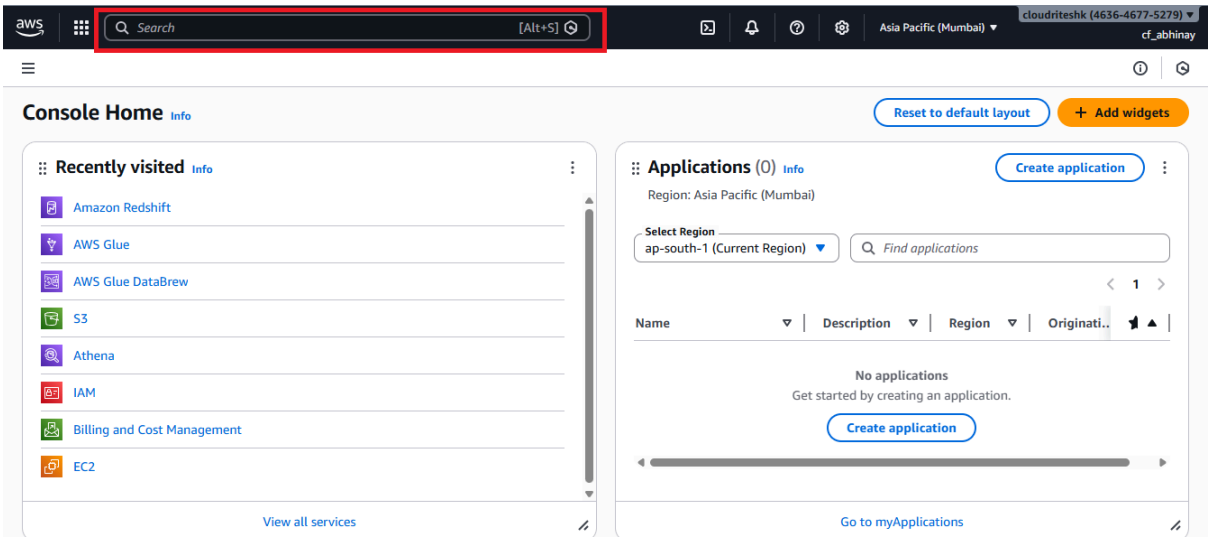
Prerequisites for the Lab

- AWS account with access to **Amazon Redshift**
- Default **VPC**, **subnets**, and **security group**
- An **S3 bucket** for IAM role creation

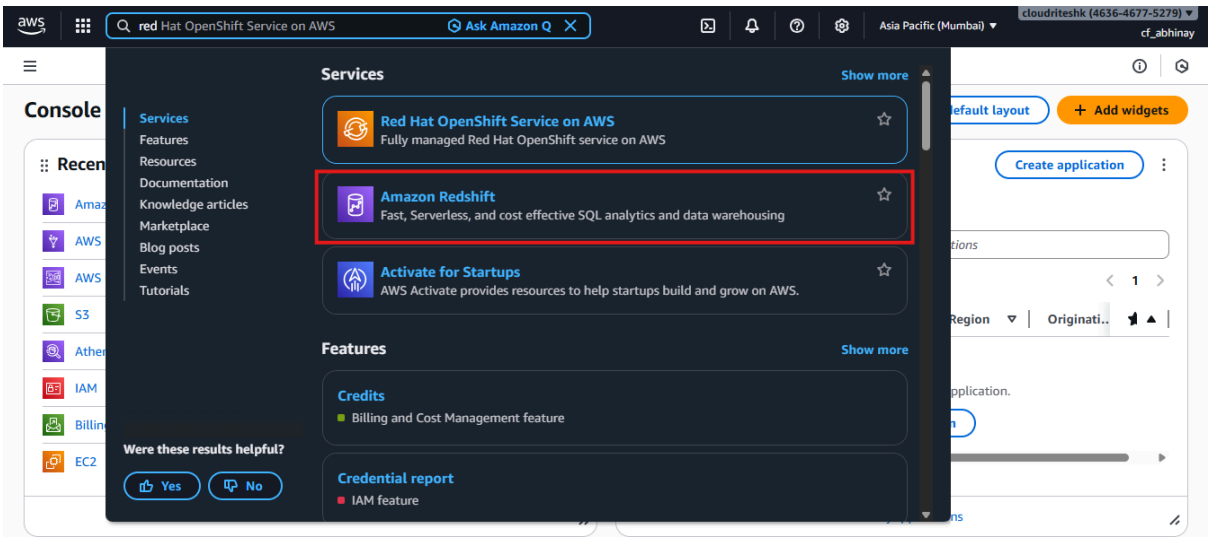
Procedure

Step 1: Navigate to Amazon Redshift

- Log in to the **AWS Management Console**

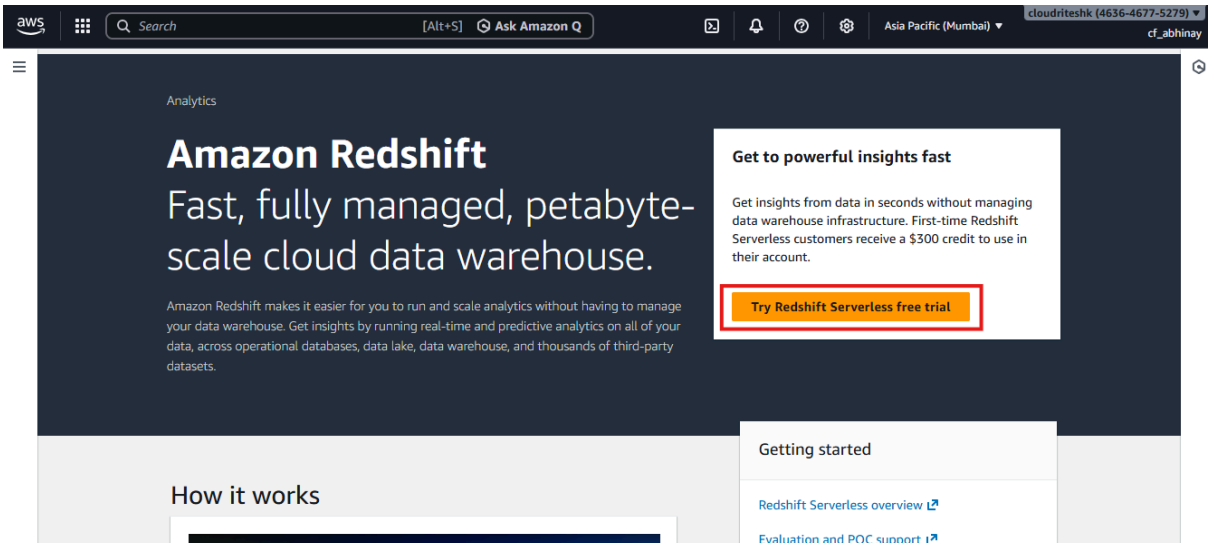


- Go to **Search bar → Amazon Redshift**

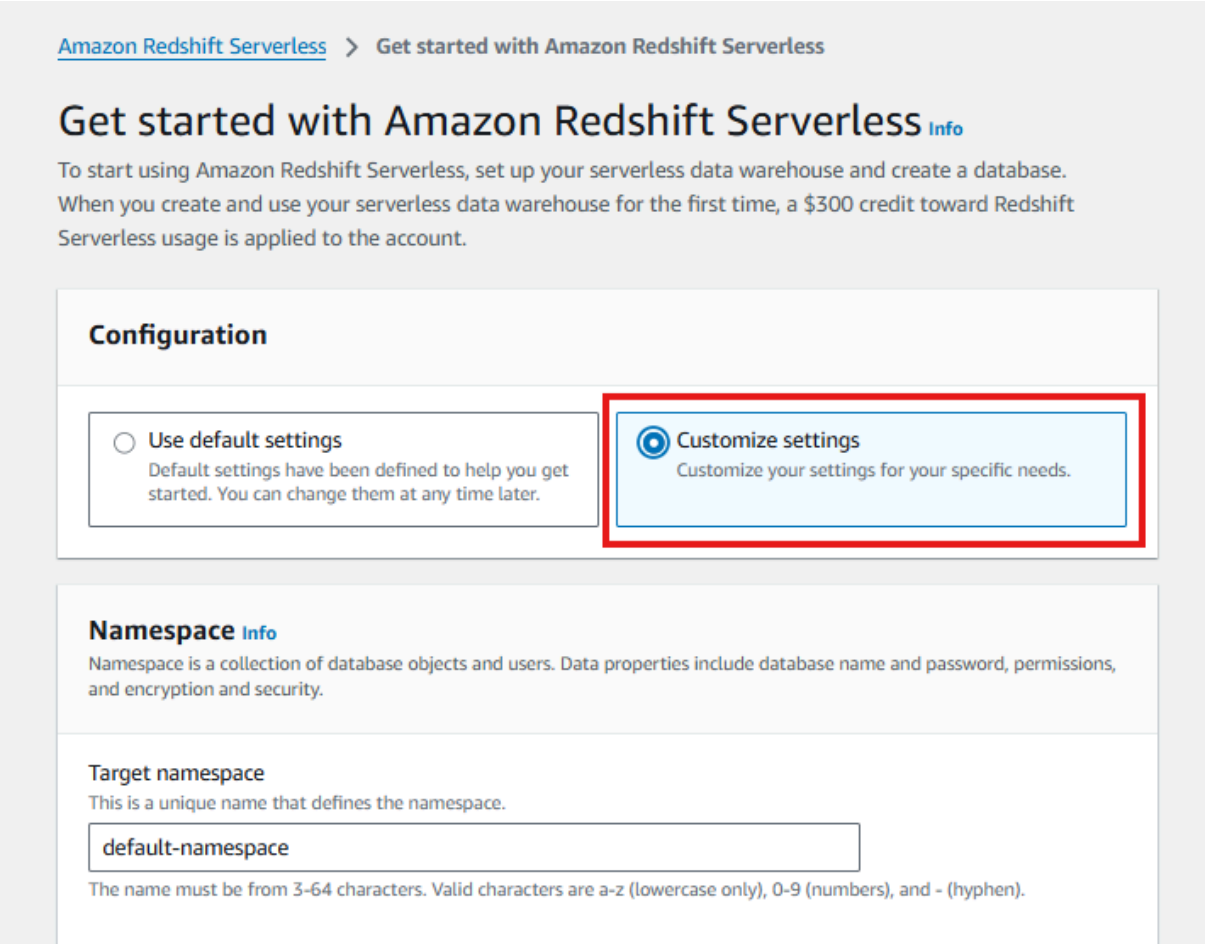


Step 2: Start Redshift Serverless Creation

- Click on **Try Redshift Serverless free trial**

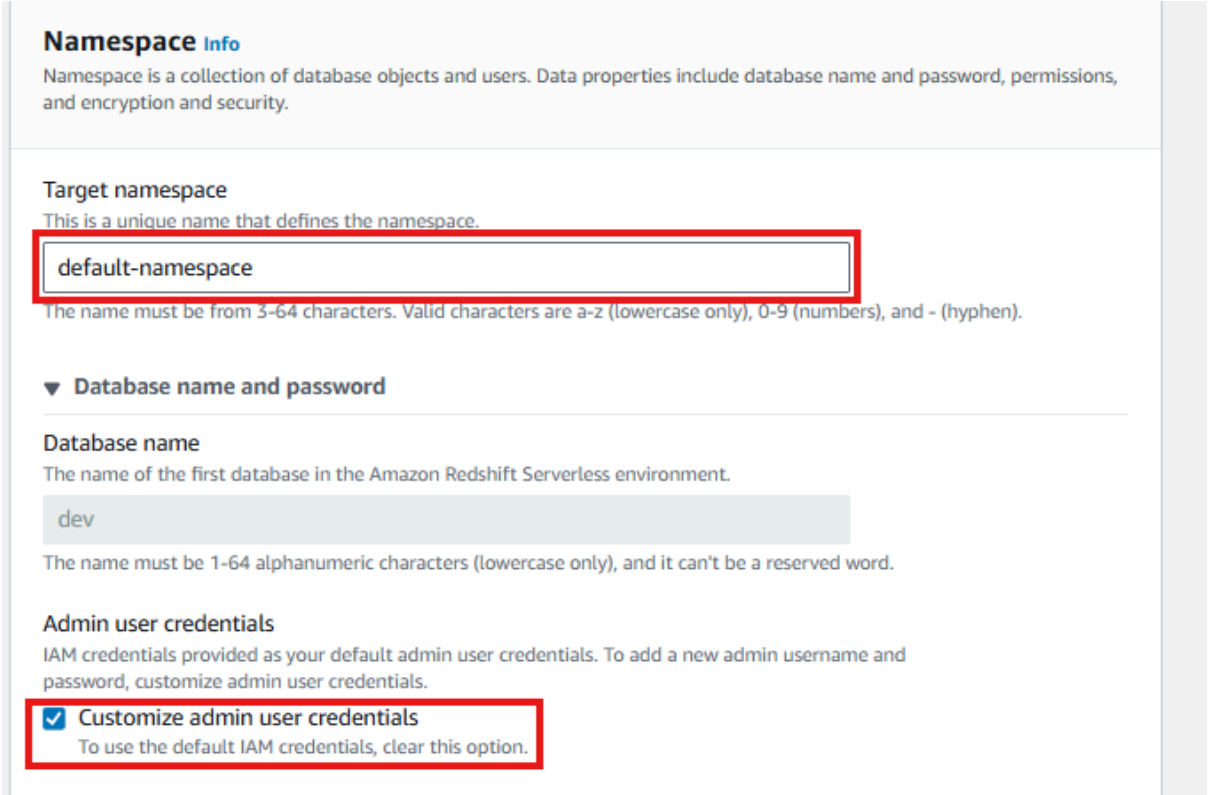


- Under **Configurations**, select **Customize settings**



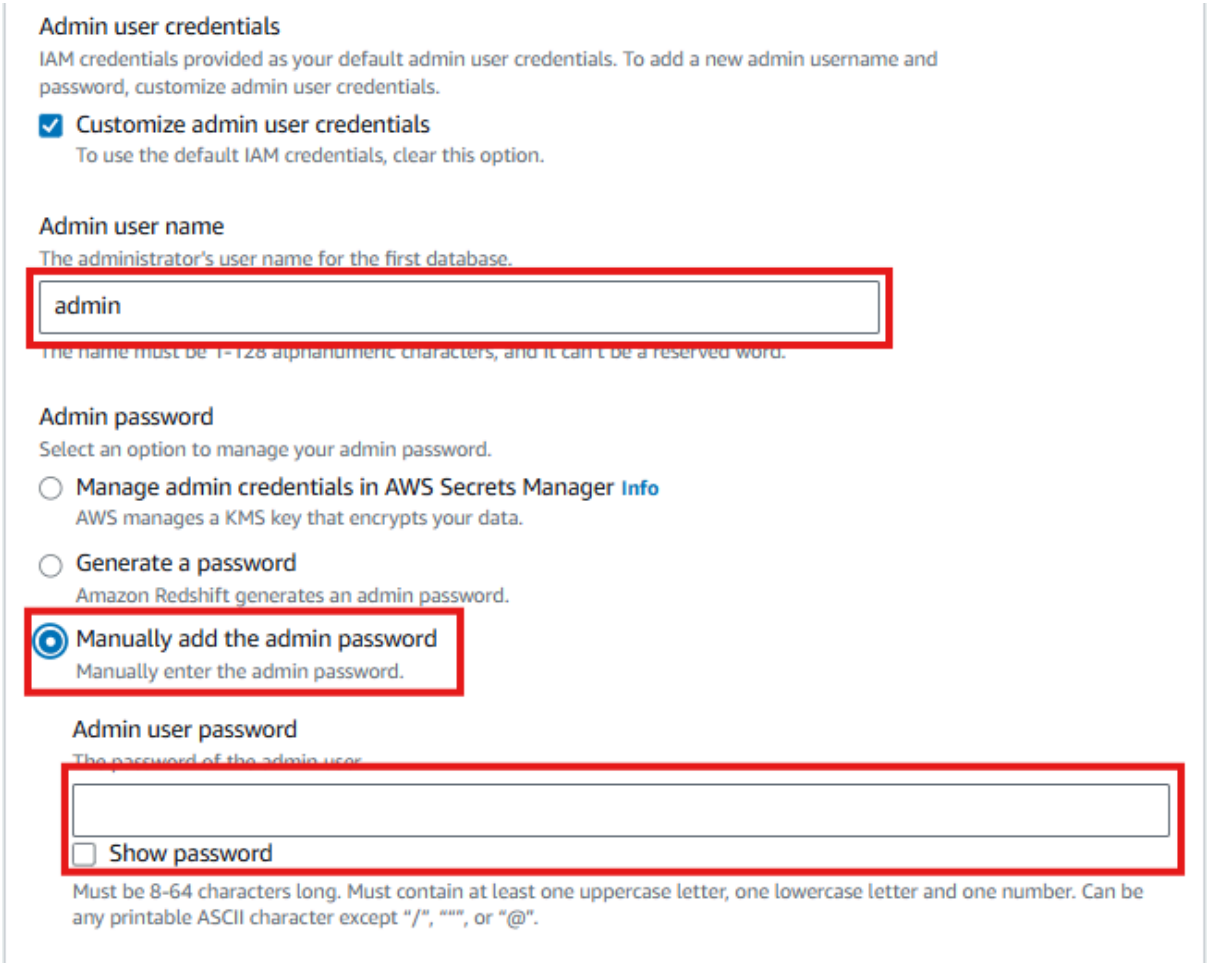
Step 3: Configure Namespace

- Leave the default namespace name as **default-namespace**
- Enable **Customize admin user credentials**

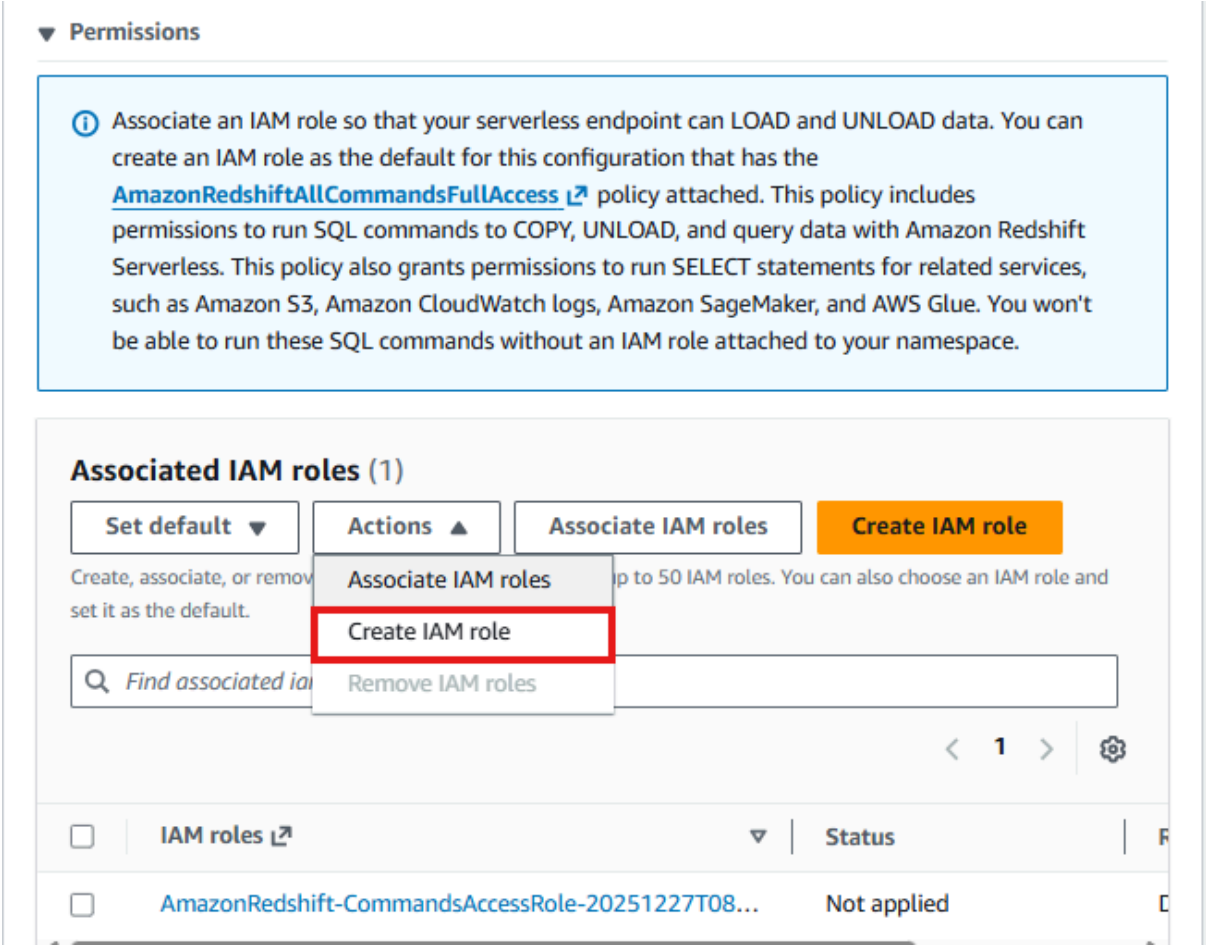


- Select **Admin user name → admin**
- Select **Admin password → Manually add the admin password**

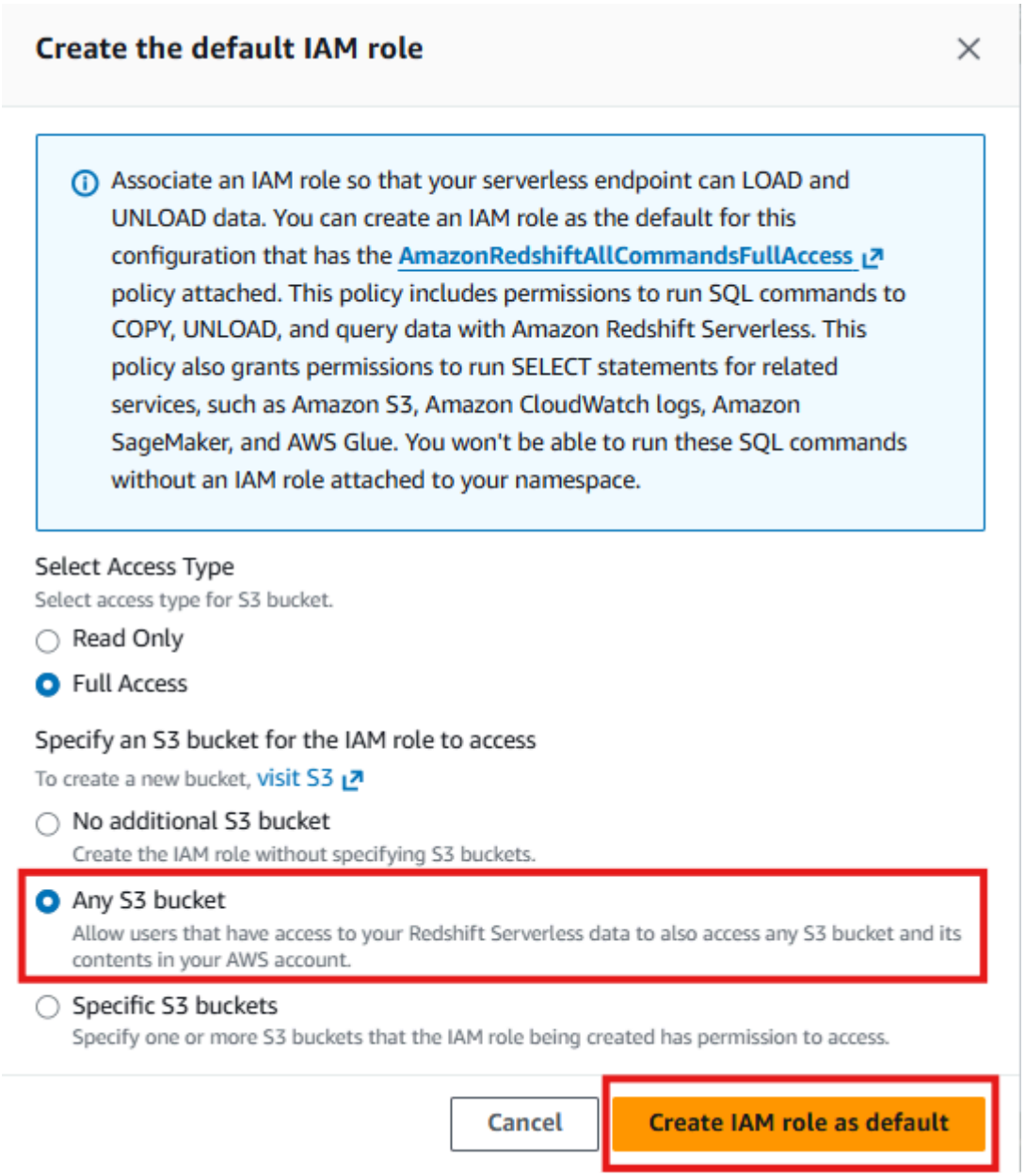
- In **Admin user password** → Enter password(remember it)



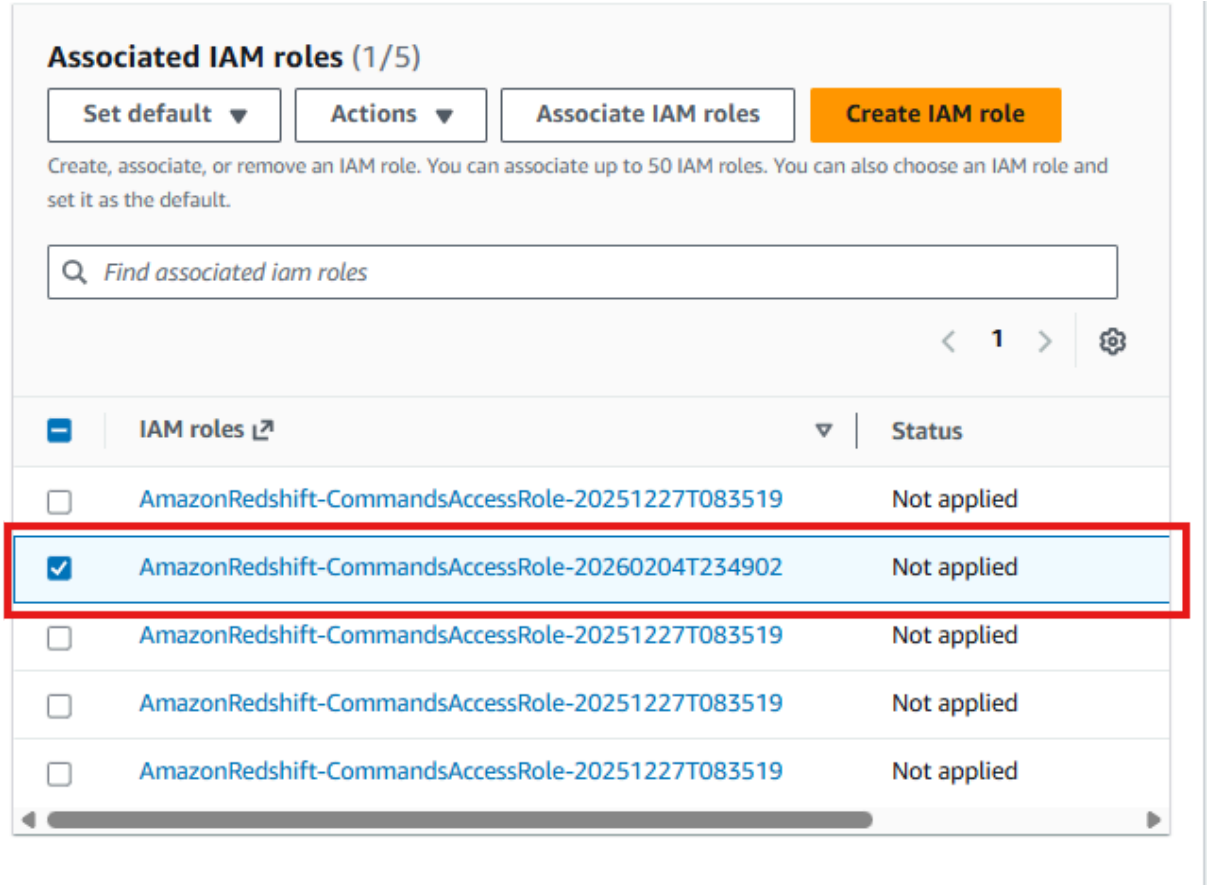
- Under **Permissions**, select **Create IAM role**



- Choose any S3 bucket and click **Create IAM role**

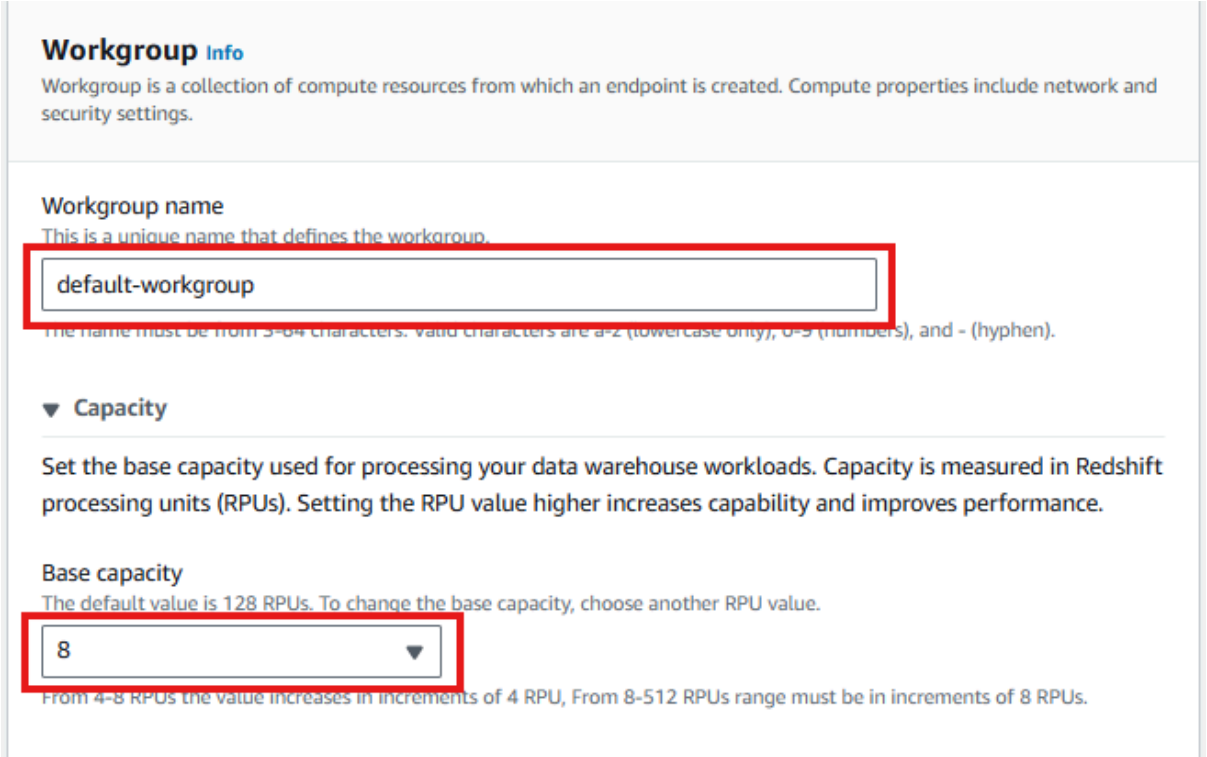


- Select the newly created IAM role



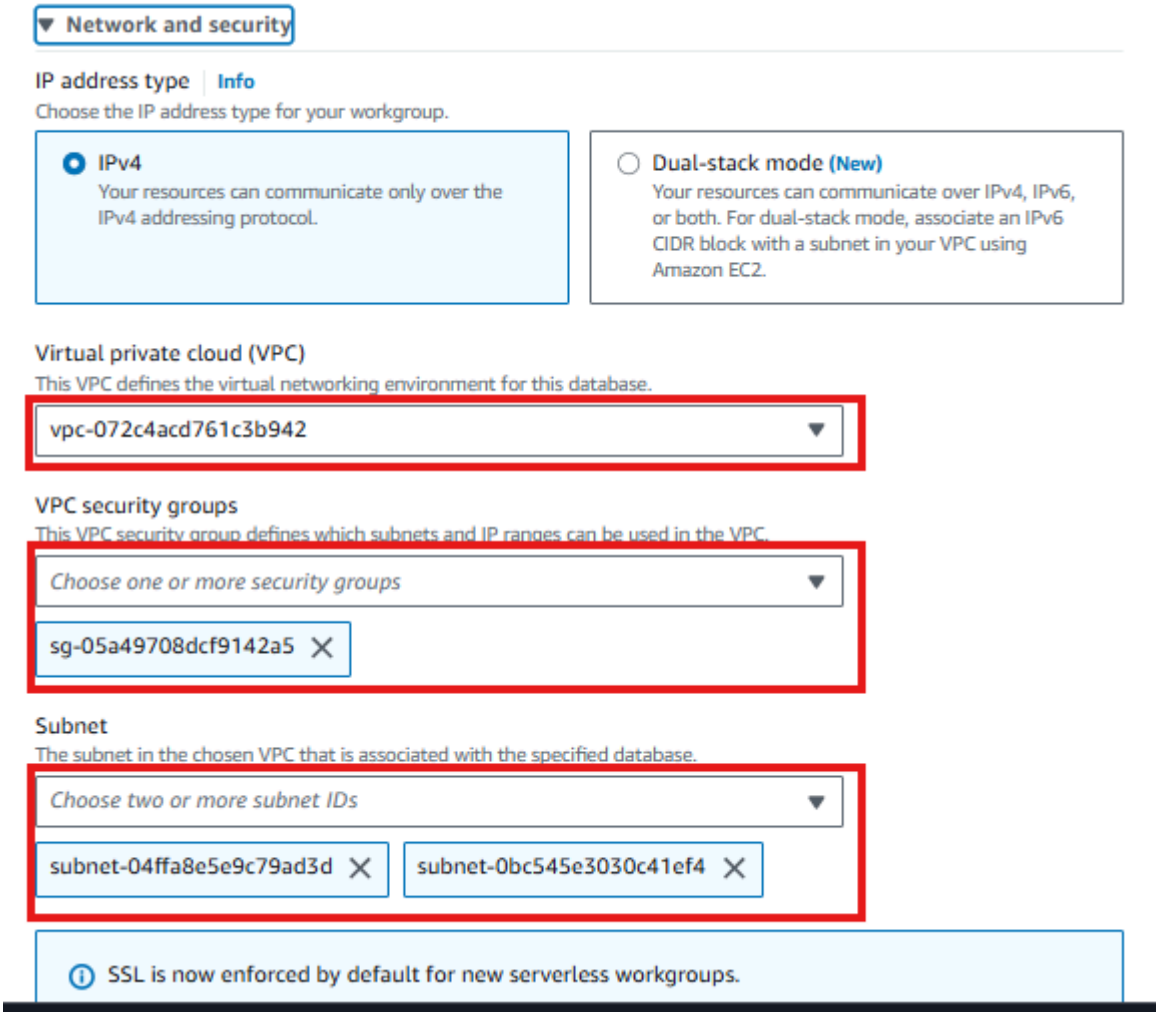
Step 4: Configure Workgroup

- Leave the default workgroup name as **default-workgroup**
- Under **Capacity**, select **8 RPUs**
    - Each RPU provides 16 GB memory



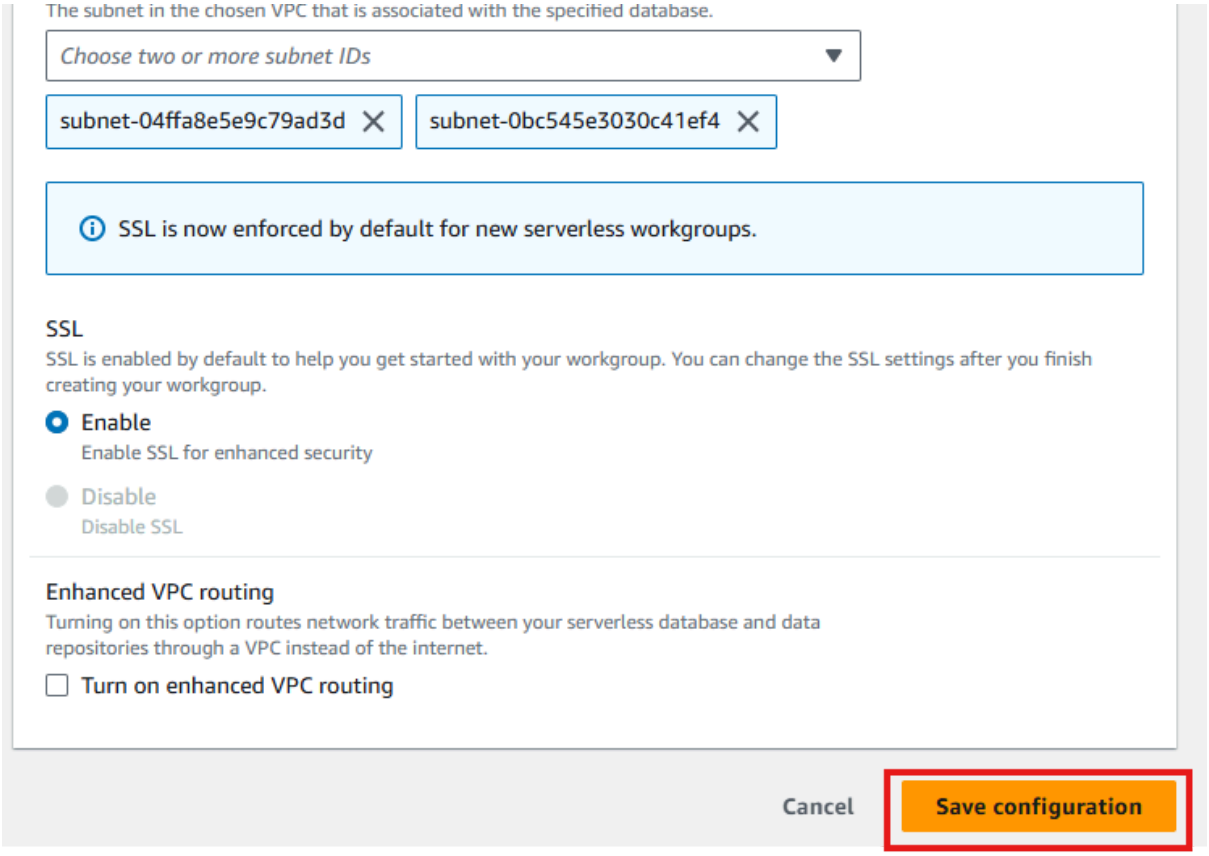
- Leave the default **VPC**, **Security Group**, and **Subnets** selected

- Ensure all subnets are selected if not auto-populated



Step 5: Save Configuration

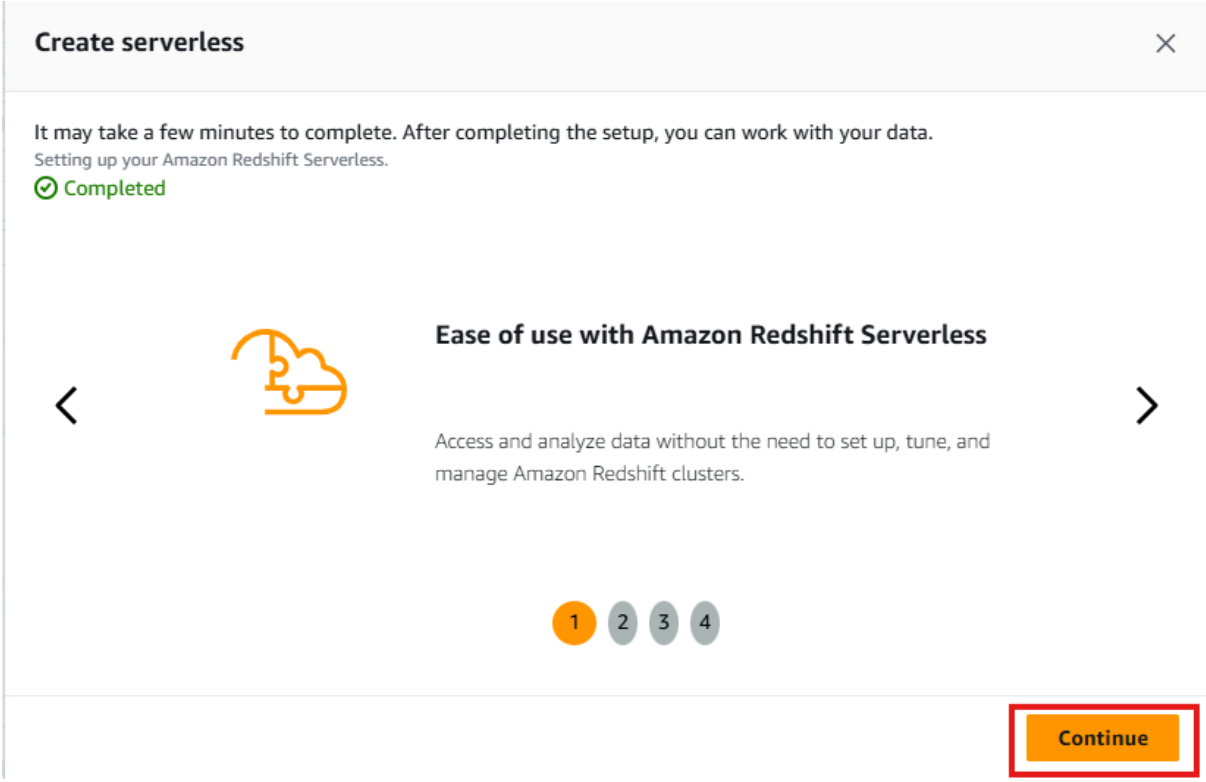
- Click on **Save Configuration**



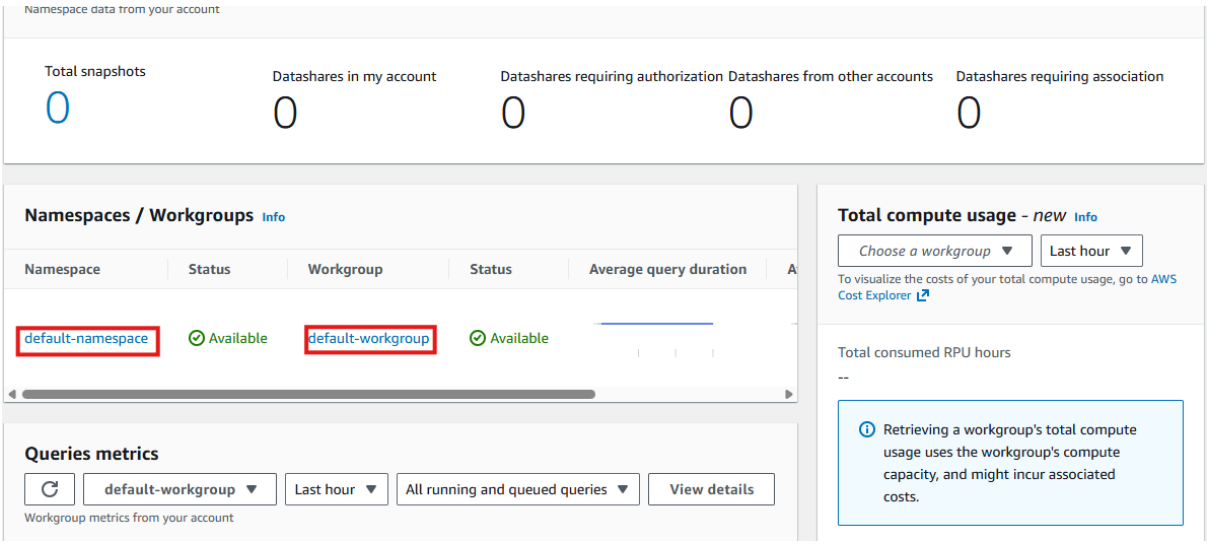
- Wait for a few minutes until the Redshift Serverless environment is created

Step 6: Verify Namespace and Workgroup

- Click **Continue** after creation

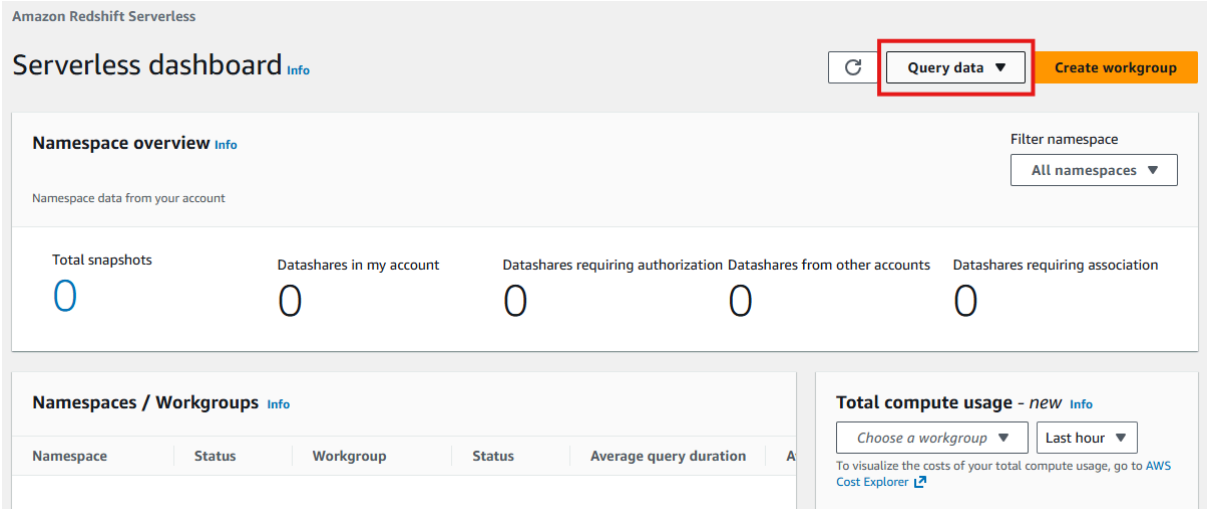


- Reload the Redshift page
- Verify that **default namespace** and **default workgroup** are visible

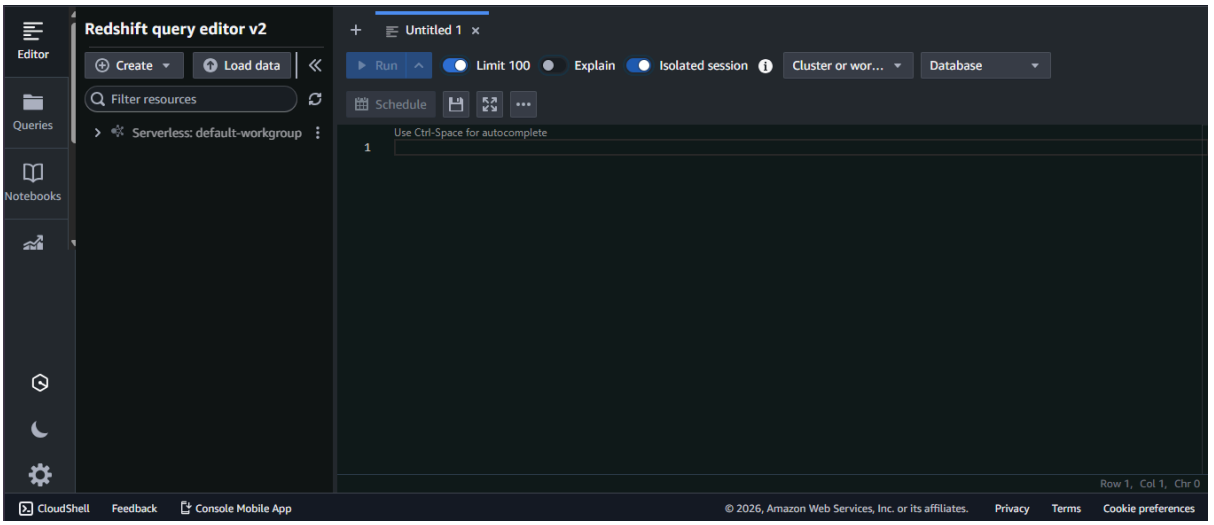


Step 7: Open Query Editor

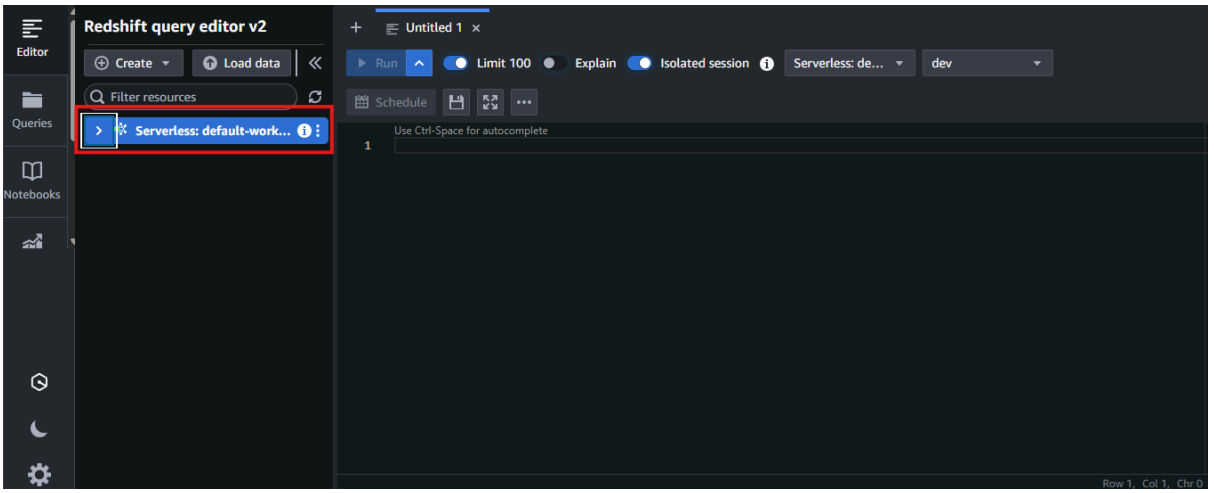
- Click on **Query data** (top-right corner)



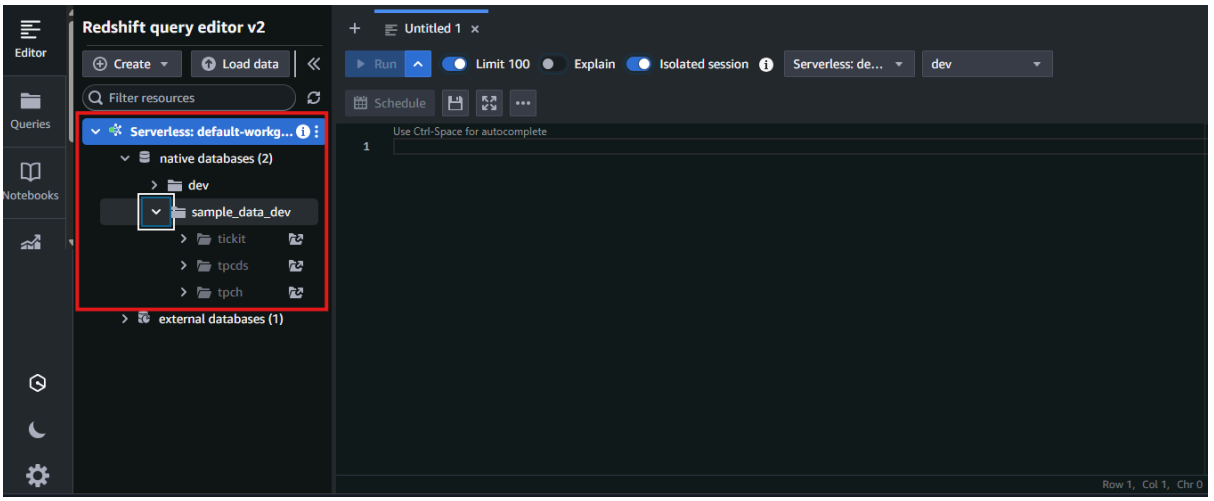
- This opens **Query Editor v2**



- Expand **default workgroup** on the left panel

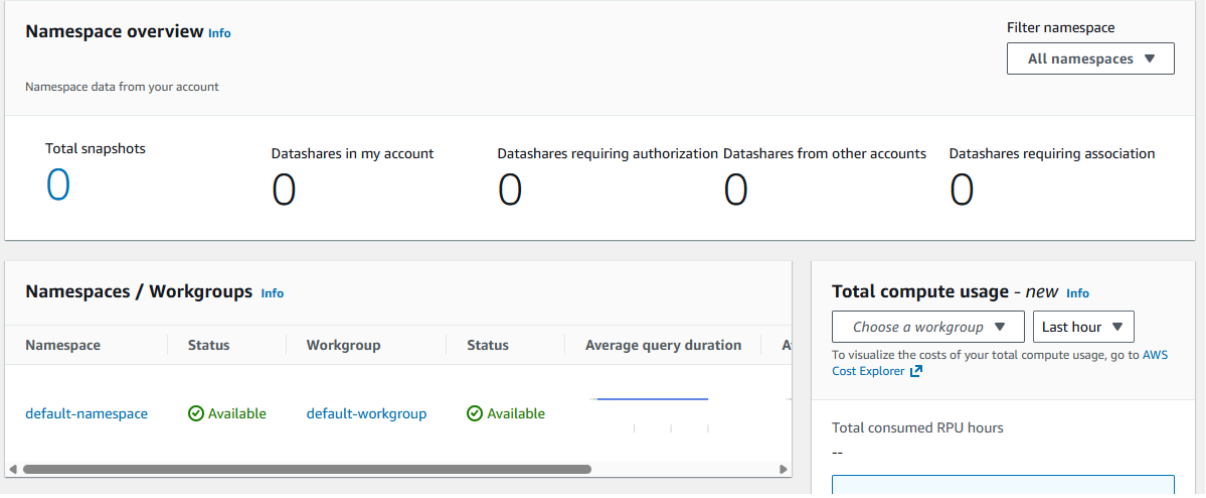


- View sample databases such as **sample_data** and **dev**

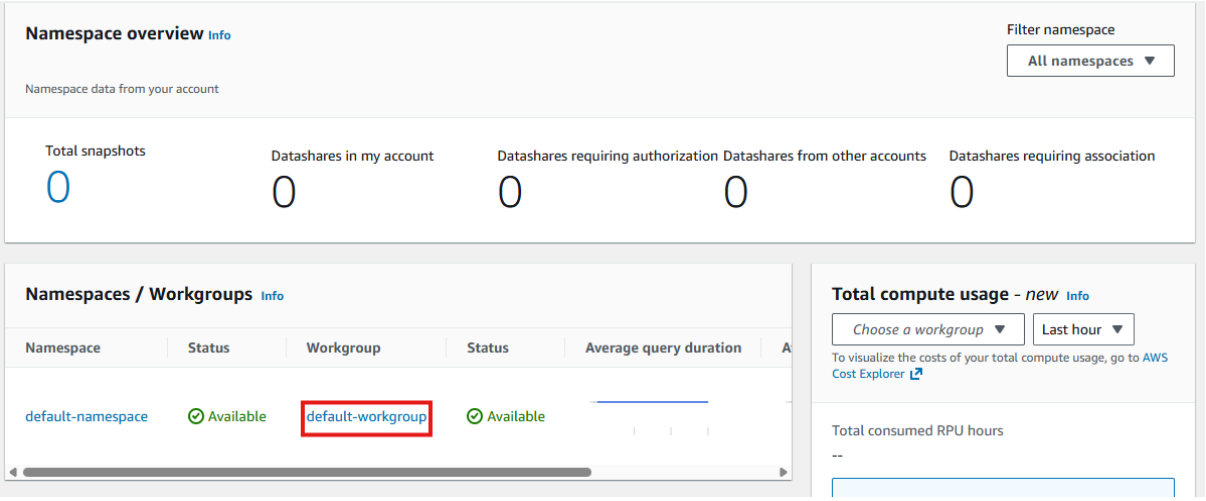


Step 8: Enable Public Accessibility

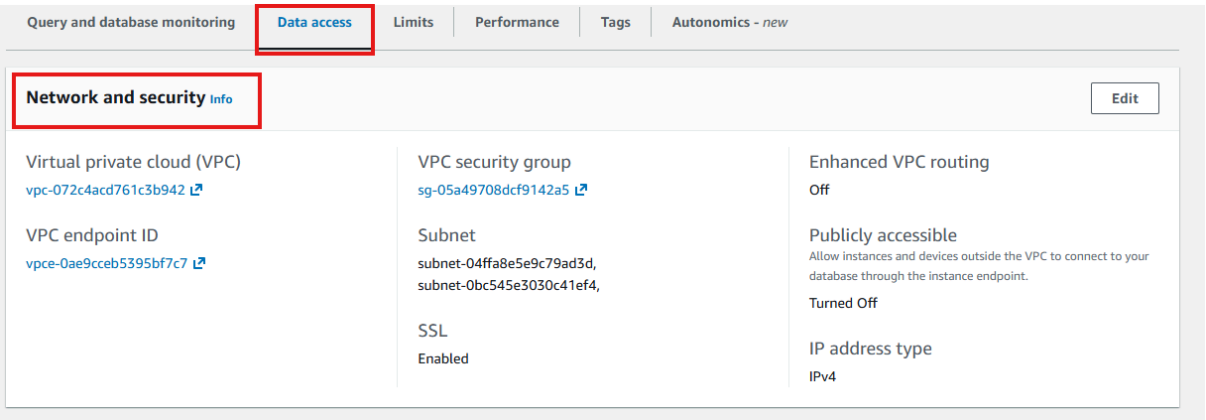
- Go back to **Amazon Redshift Console**



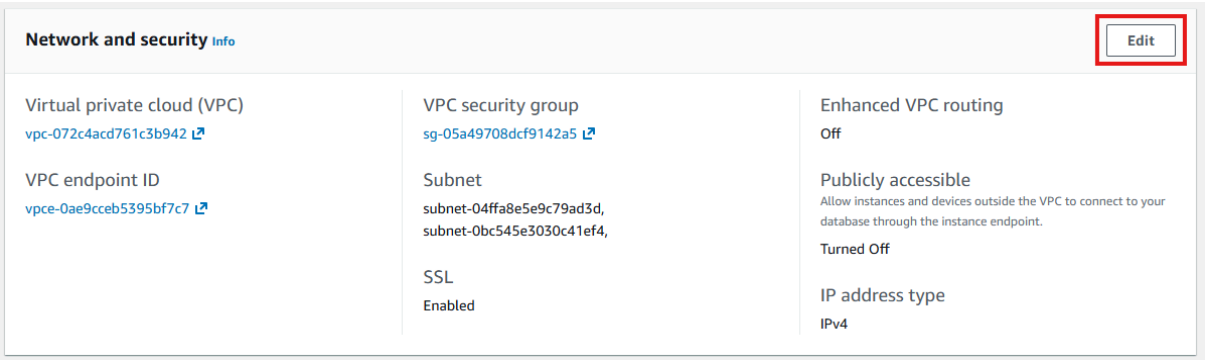
- Click on the **Workgroup**



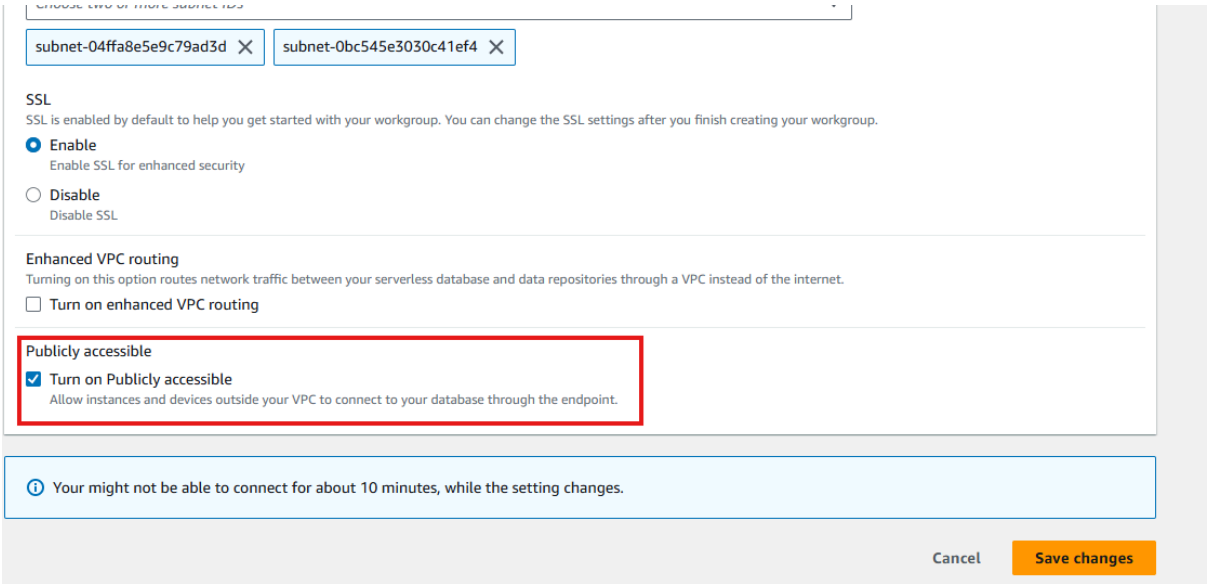
- Navigate to **Networking and Security → Data Access**



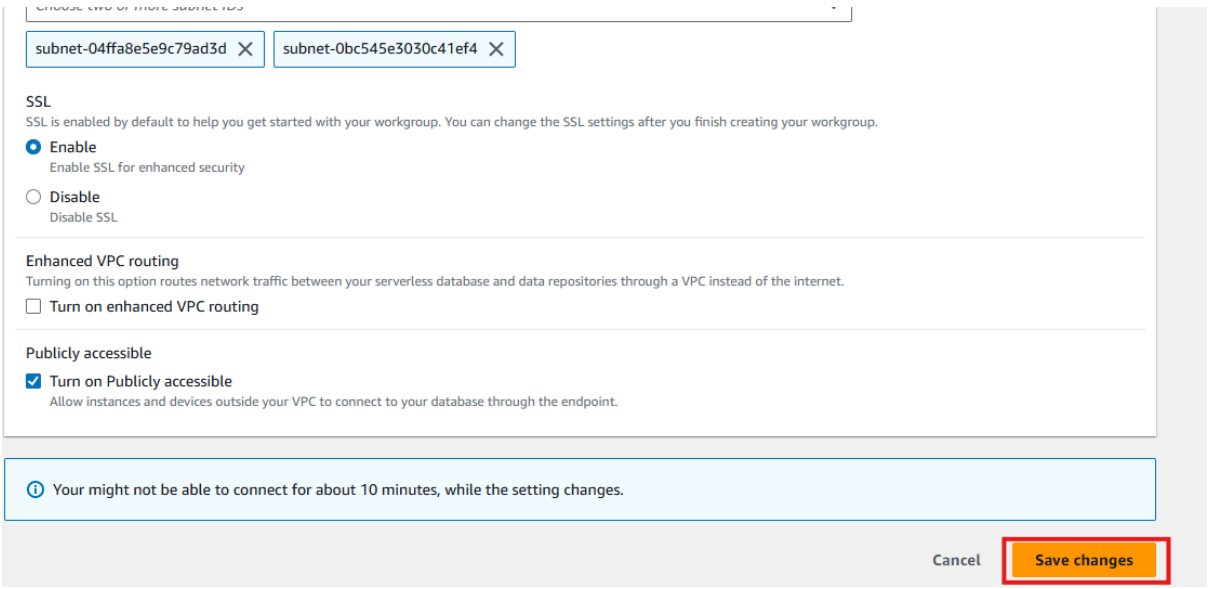
- Click **Edit**



- Enable **Turn on publicly accessible**

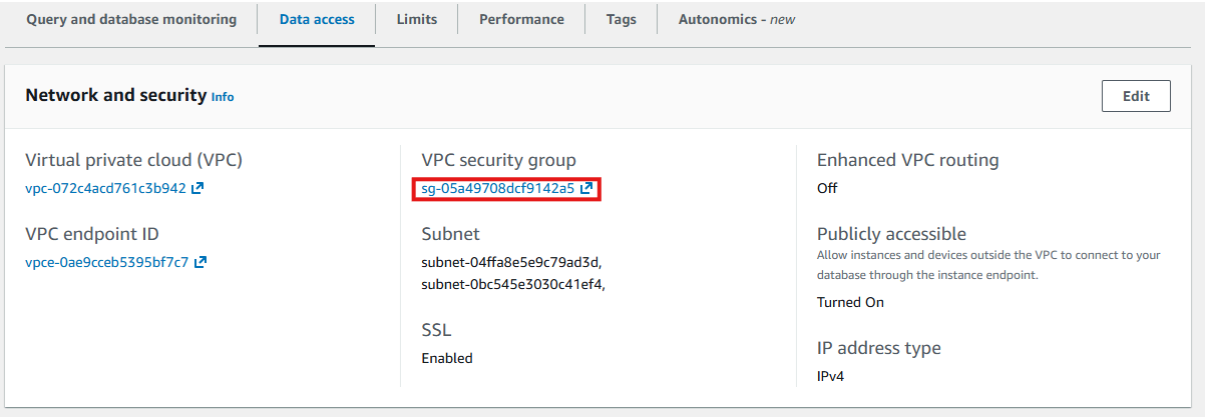


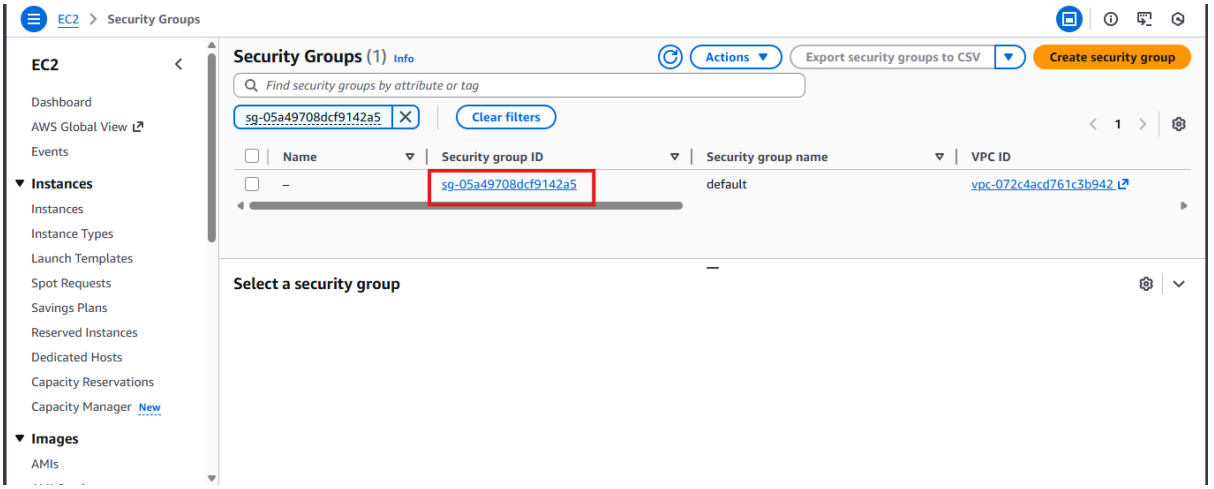
- Click **Save changes**



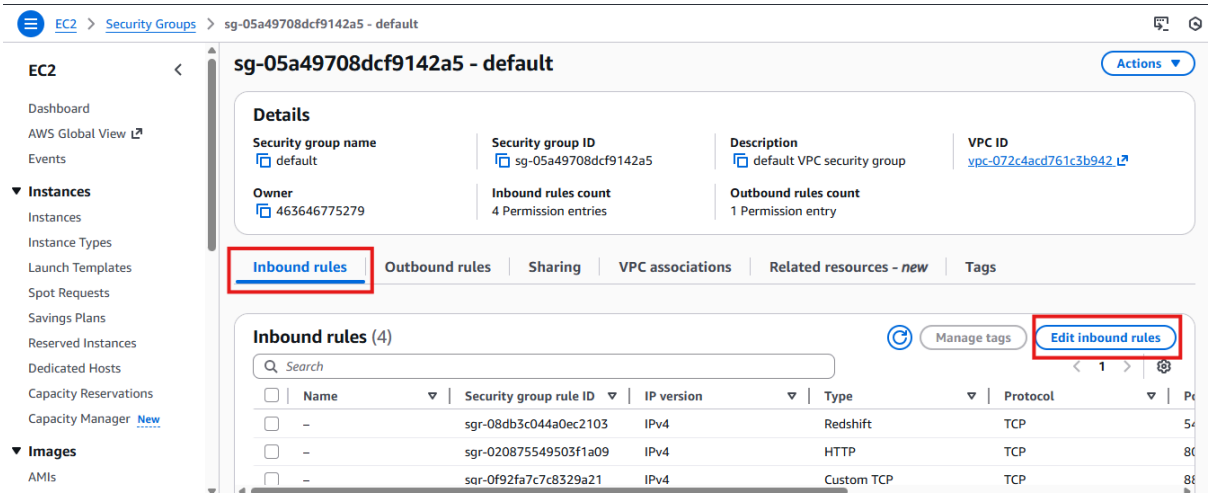
Step 9: Configure Security Group Inbound Rules

- Under **Networking and Security**, click on the **Security Group**

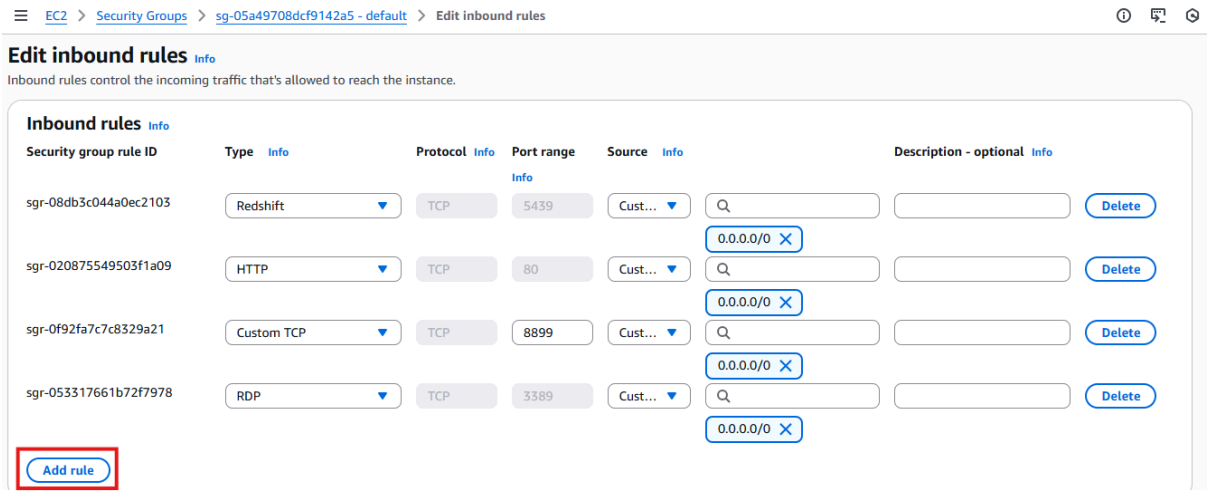




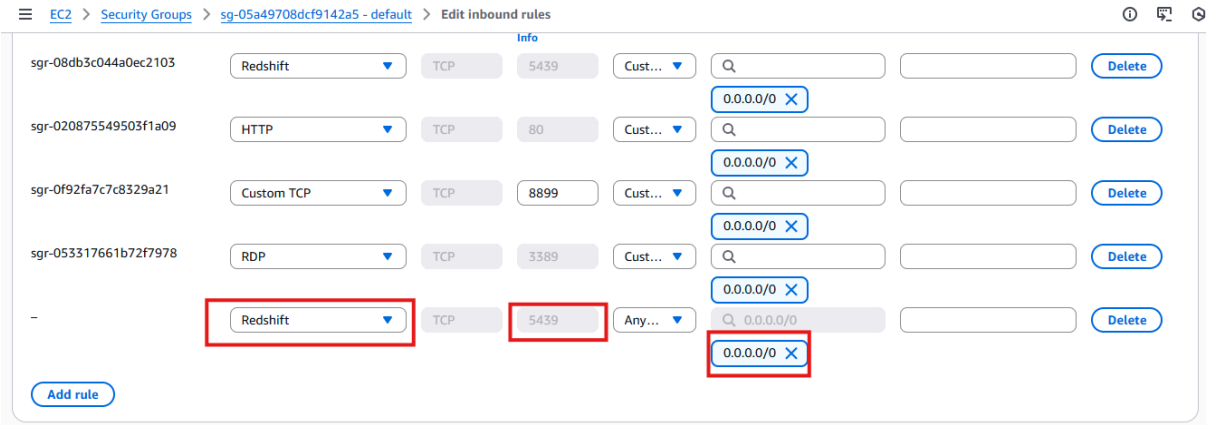
- Select **Edit inbound rules**



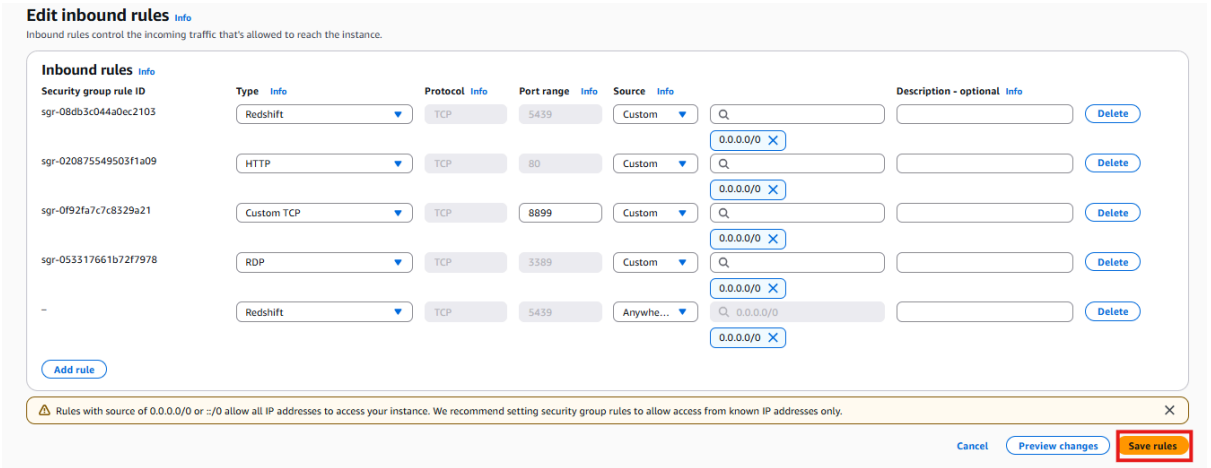
- Click **Add rule**



- Configure:
  - **Type:** Redshift
  - **Port:** 5439 (auto-populated)
  - **Source:** 0.0.0.0/0



- Click **Save rules**



Note: Allowing 0.0.0.0/0 is not recommended for production but is acceptable for lab purposes.